

113B Data Science and Python Programming

Homework Assignment #2

Due: 2025/4/30 12:00:00

This assignment focuses on applying data science and Python programming techniques to simulate the corrosion behavior of natural gas pipelines using the Monte Carlo method and assess the risk of gas leaks. By modeling environmental corrosion conditions and internal pressure loads using random variables, we will analyze the impact of different corrosion control strategies on pipeline life and conduct hypothesis testing using statistical methods to verify the significance of control quality on leakage probability. This not only helps to understand data-driven engineering decisions, but also provides a basis for practical application in reliability assessment.

Problem #1: Monte Carlo Method (40%)

Consider a buried natural gas pipeline segment with the following properties:

Length	Outer Diameter	Wall Thickness	Steel Alloy
10 km	1 m	12.3 mm	API 5L X70

During operation, the pipe segment is expected to encounter corrosive conditions. According to the historical experience, the expected corrosion rates (in millimeter per year) X in different environmental and corrosion control conditions will follow an exponential distribution.

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

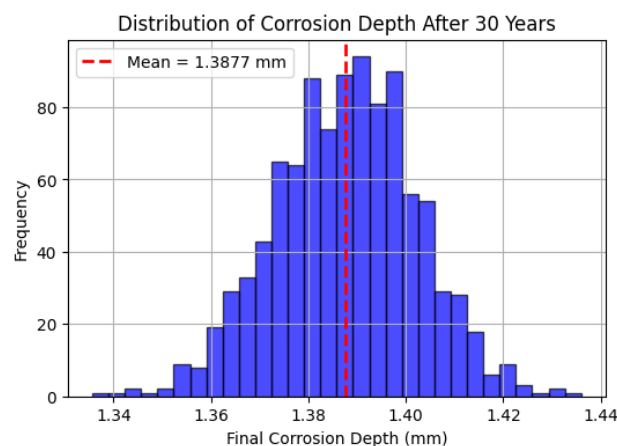
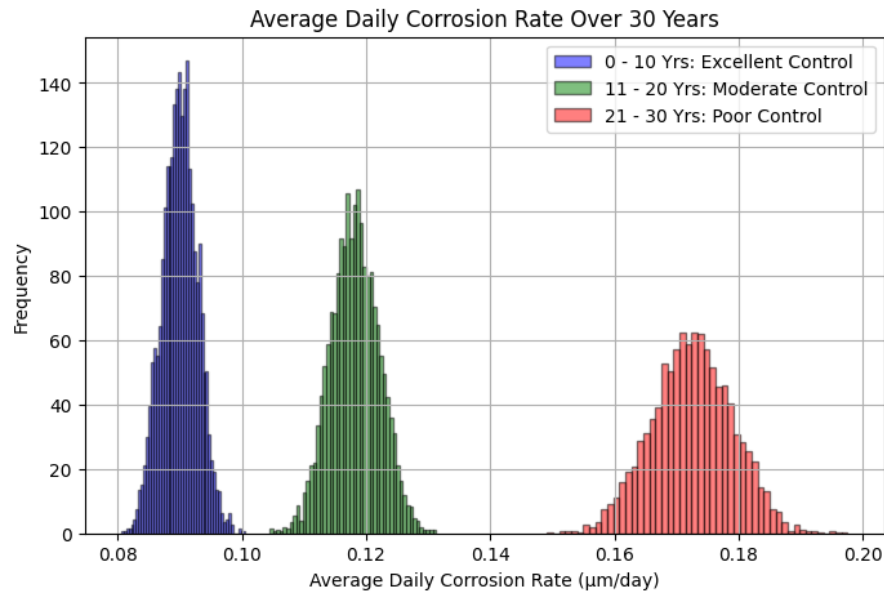
Let us assume that we already know the hyperparameter λ in different environmental and corrosion control conditions as shown in the following table.

	Very Aggressive	Moderately Aggressive	Mildly Aggressive
Poor Control	9.61	20.34	27.95
Moderate Control	17.22	24.91	32.05
Excellent Control	25.18	29.35	40.35

Assuming **the environment aggressiveness can change randomly from day-to-day**, and **the quality of the corrosion control is excellent at the start, moderate after 10 years, and poor after 20 years**. Please write a program to **simulate the corrosion**

depth (D) in millimeter over 30 years of operation. This result should be a distribution of corrosion depth D . Also, please report a 95% confidence interval for the average corrosion depth D over 30 years.

If everything goes well, you might have these results:



[???, ???] is approximately a 95% confidence interval for the average corrosion depth after 30 years of operation.

FRIENDLY REMINDER

You don't need to follow the figure style.

Feel free to create your own style!

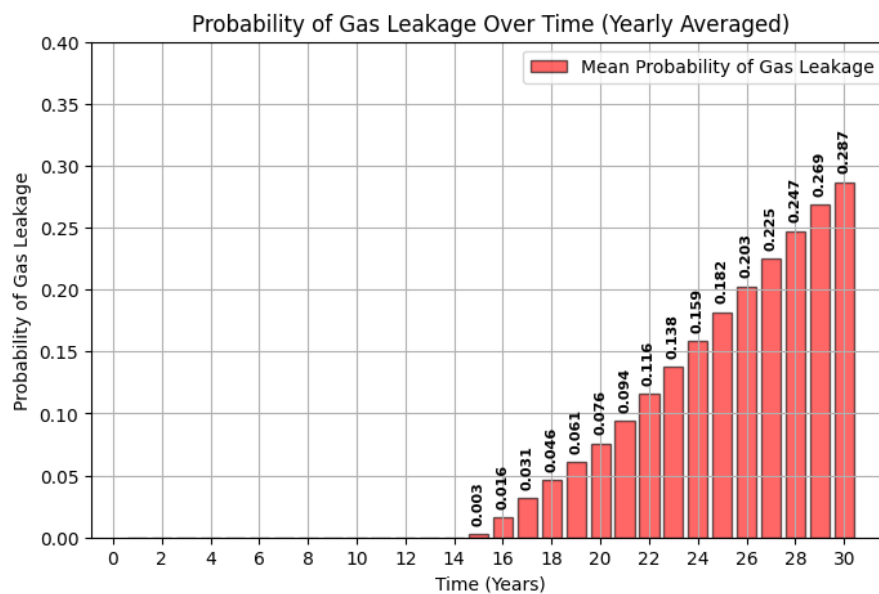
Problem #2: The Chance of Gas Leak (40%)

Operationally the pipe can go through random internal loading pressure once per day. The internal loading pressure value P is uniformly distributed between 6 and 8 MPa. For any given internal pressure, the pipe can fail if the pipe thickness has reached a minimum value $T_{\min}$ as result of corrosion thinning of the wall. Assume that the minimum thickness $T_{\min}$ for safe operation is defined as follows,

$$T_{\min} = \frac{P_{\max} \cdot D_{\text{out}}}{1.4 \cdot Y_s},$$

where $P_{\max}$ is the maximum internal loading pressure, D_{out} is the outer diameter of the pipe, and Y_s is the yield strength of the pipe (for API 5L X70, $Y_s = 485$ MPa). Using the simulation you have done in problem 1 (Monte Carlo simulation for the corrosion depth over time with the operating conditions of the pipe), **what is the expected probability of gas leakage by year of operation?**

If everything goes well, you might have these results:

**FRIENDLY REMINDER**

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Feel free to create your own style!

Problem #3: Hypothesis Checking (20%)

Suppose we want to examine the effect of corrosion control quality on the risk of gas leakage in a pipeline. Use Monte Carlo simulation results to test the following hypotheses:

1. H_0 : Corrosion Control Quality has no significant effect on the probability of gas leakage.
2. H_1 : Corrosion Control Quality has a significant impact on the probability of gas leakage.

Hint: You can simulate the mean corrosion depth under different quality controls over 30 years. And compare their difference as your test statistic. In this case, your test statistic will be

$$W = \frac{\mu_1 - \mu_2}{S_p \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where

$$S_p = \sqrt{\frac{(n_1 - 1) \cdot S_1^2 + (n_2 - 1) \cdot S_2^2}{n_1 + n_2 - 2}}$$

What is the P -value of your simulation that results in rejecting H_0 ? Can we reject the null hypothesis based on a level α test? Please return a corresponding message in your program.

Here is the sample code for your main script:

```
if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="Homework #2")
    parser.add_argument("--n", "--num", default=1000, type=int,
        help="Number of trials (default: 1000)")
    parser.add_argument("--y", "--years", default=30, type=int,
        help="Number of years (default: 30)")
    parser.add_argument("--a", "--alpha", default=0.05, type=float,
        help="Level of confidence (default: 0.05)")
```

剩下你自己想喔

Please accomplish this homework with an organized code (e.g., with main script and function script). Here is a template for your code structure:

```
113B_hw2_0123456789
├─ ans.py          # Solution of homework #2
├─ xxx.py          # xxx module if necessary
└─ main_hw2.py     # Main script of homework #2
```

You don't need to follow this structure, just keep your main script clean.

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs (e.g., signature). **Your signature should be attached in your main script.** Here is the template of signature:

```
"""
Created on Wed Aug 7 01:23:45 2024

@author: Xi Winnie, student ID

@collaborators: William Lai, his student ID
"""
```

Please save your code as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

113B_hw2_**ID**.zip

For example,

113B_hw2_**0123456789**.zip

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “E3@NYCU” system. There will be a section for uploading your homework.